Bilkent University EE313 ECD Lab 1 Work Report

Diode Characterization and Differential Temperature Sensor

Özgür Ekin Sönmez [ID]
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Introduction and Aim

In this lab, firstly our aim is to find the Is of a p-n diode with a given diode current formula. After that in part B, our aim is to create a differential temperature sensor with given specifications by using a diodes voltage dependance of temperature. At first, these procedures are done by simulation then on hardware. A comparison will be made to make sure that our experiment worked properly.

SOFTWARE PART:

Part A:

In this part, we were required to show and measure the I_S of a p-n diode.

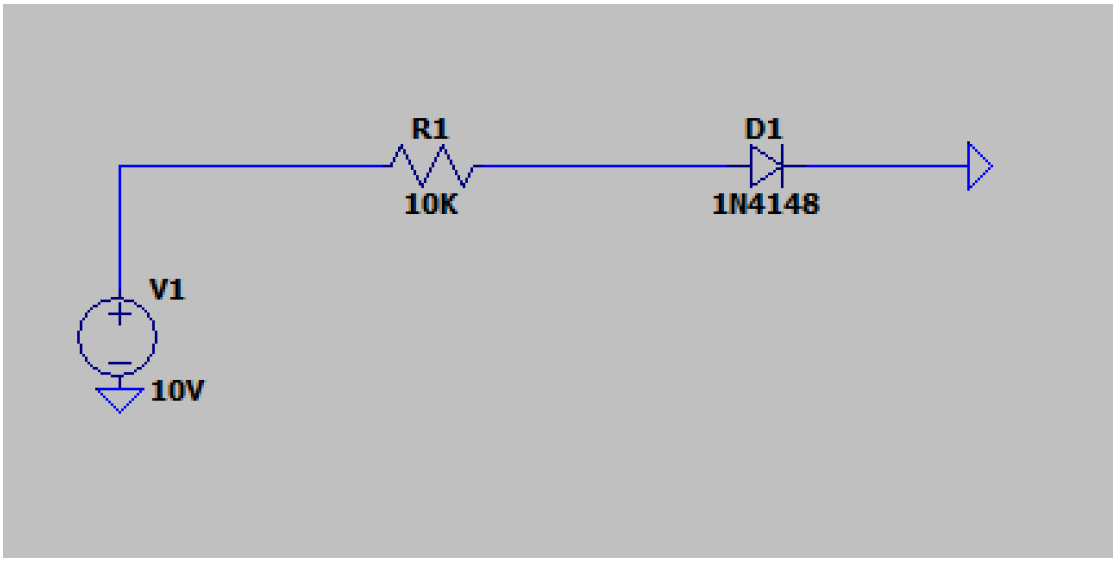


Figure 1.0: Circuit created with R= 10KΩ and Vcc=10V

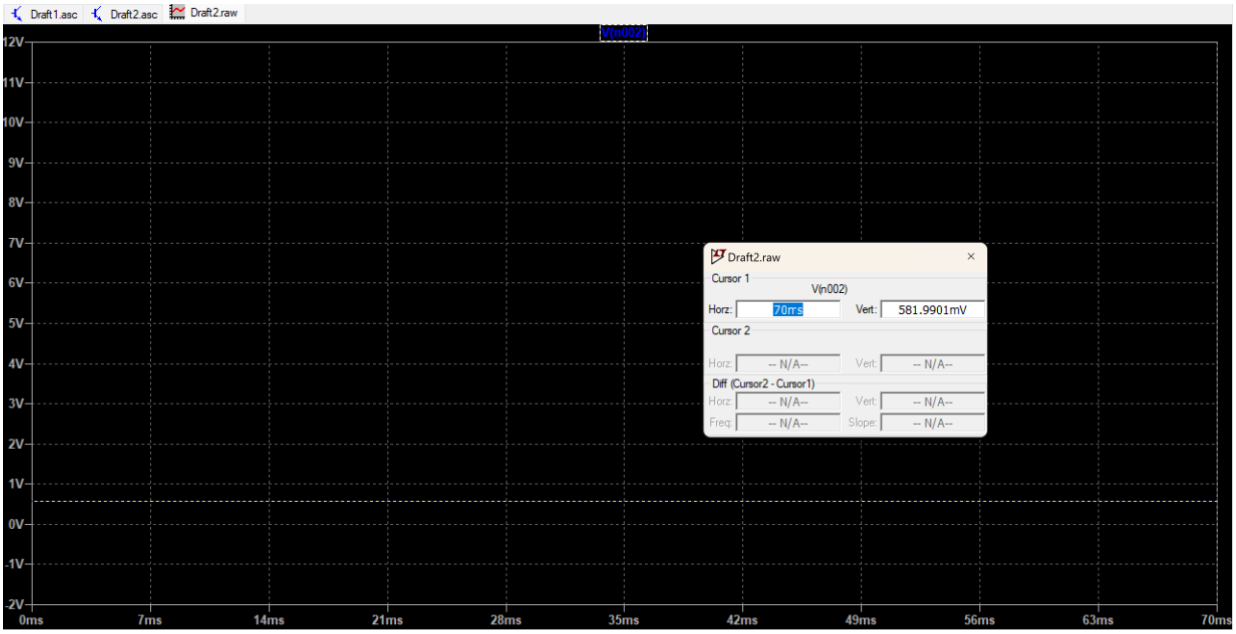


Figure 1.1: Vd value.



Figure 1.2: Is value.

As it can be seen from the figures the Vd value is 581.99mV and the Id value is around 941.8 µA

After these measurements, we can simply calculate the Id value according to this formula:

$$I_d = I_s * (e^{\frac{q*V_d}{n*k*T}} - 1)$$

with q=1.6*10^-19coulombs, k=1.38x10^-23J/K, T=300K, and n=1.752, we can find $I_s \approx 2.502$nA which is pretty close to the spice value of 2.52nA

Part B:

In this part, the circuit shown in Figure 2.0 is designed:

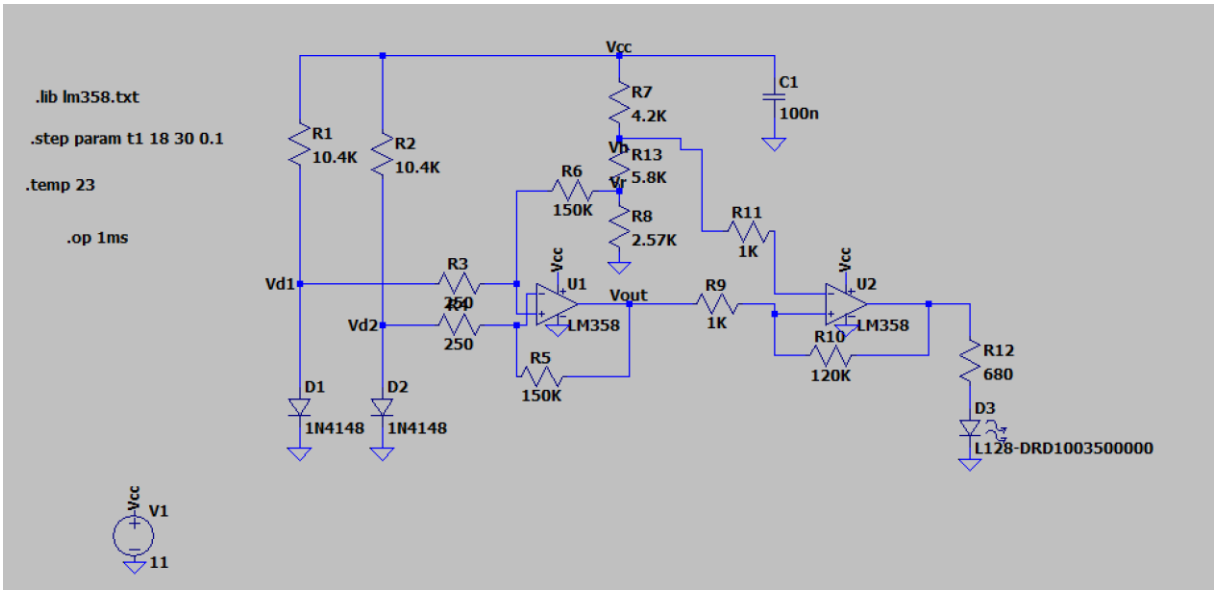


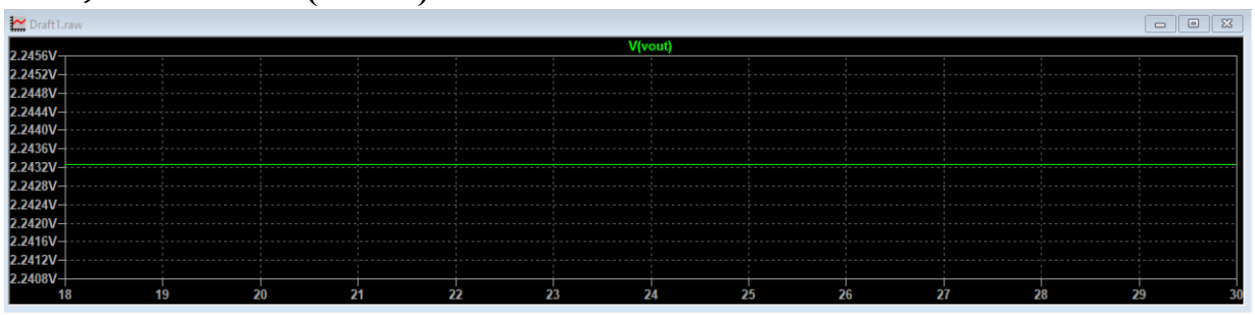
Figure 2.0: Final Circuit

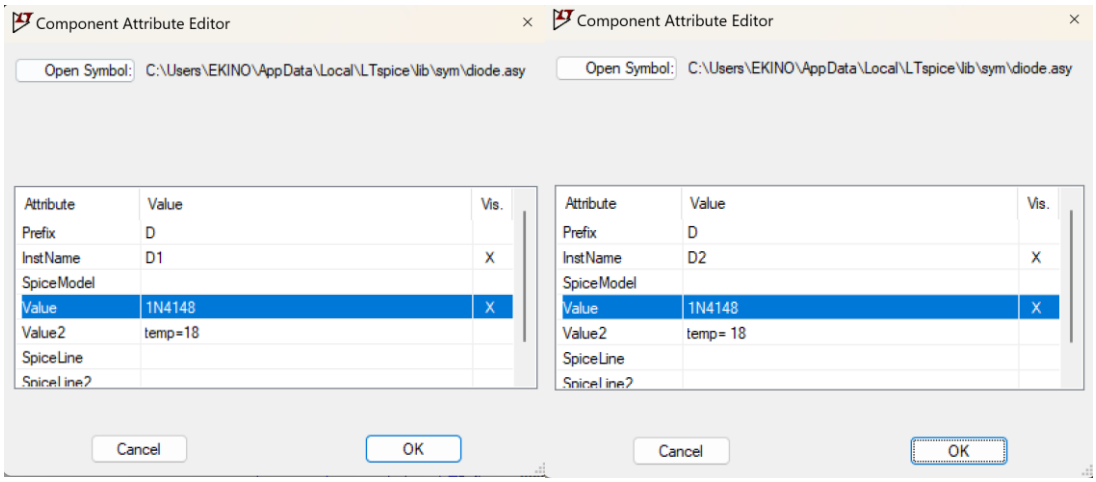
First of all, Vcc is chosen as 11V since my Bilkent ID is ending with the digit 6.

We have 2 diodes. One of them(D1) works as a reference diode for the room temperature and the other one (D2) works as a sensor diode. With each 1 degree celcius change of the temperature, the voltage on that diode changes 2mV. This is our fundamental starting point for our temperature sensor. To begin, we need to increase the ratio of this change. This process can be completed by using an amplifier OPAMP design(U1). The second OPAMP(U2) is used as a comparator OPAMP circuit. When it is +5±0.5°C larger than the room temperature in the environment, the LED is supposed to turn ON. Here are the different several requirements for this lab and their analysis shown:

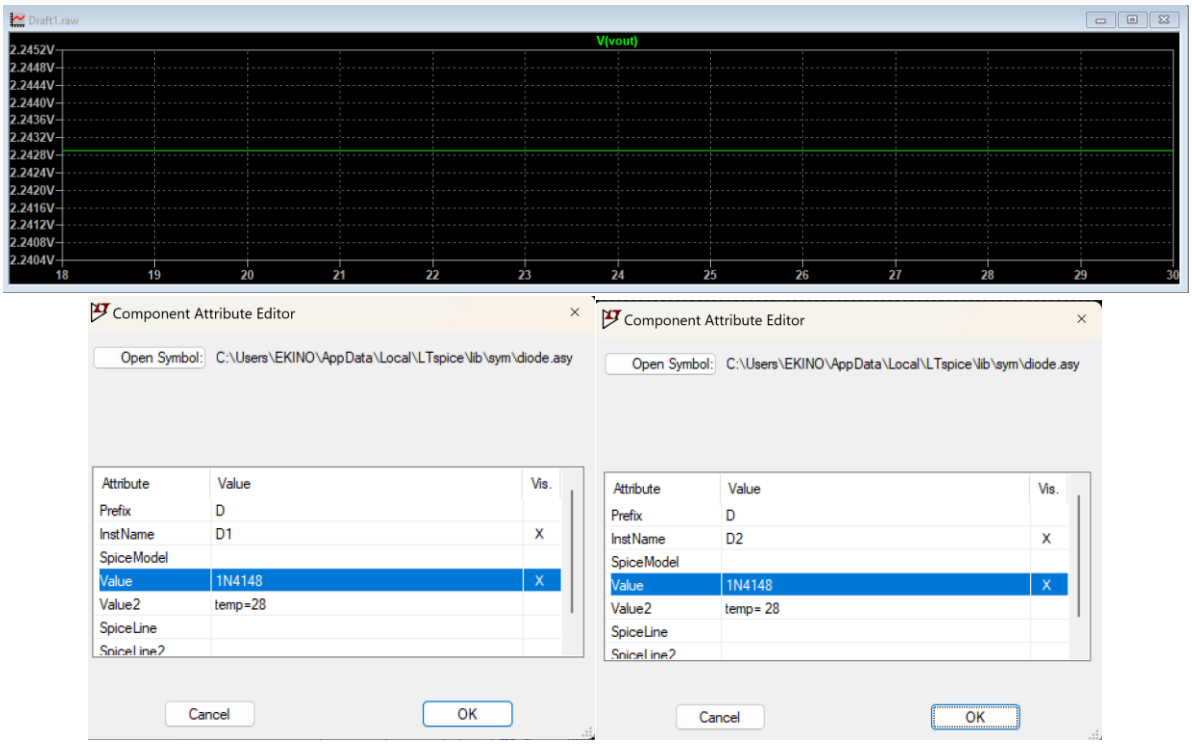
Requirements and their analysis:

1. **When both diodes are at room temperature (vD1=vD2), the output voltage, vout, should be at (Vcc−2)/4**





Figures 2.1, 2.2, 2.3: Vout with both diodes at 18 degree Celcius.



Figures 2.4, 2.5, 2.6: Vout with both diodes at 28 degrees Celcius.

As it can be seen from both of the graphs, Vout is around 2.2432V at 18 degree celcius and around 2.2428V at 28 degrees celcius which are accurate for Vcc=11V.

2. **The output voltage should show the temperature difference between the ro om temperature and the temperature of the sensor diode in degrees with a 10% tolerance. For example, if the sensor diode is +1-**
   degree warmer, vout should change by +1V.

When the room temperature is 23 degree celcius, the results can be seen as shown:

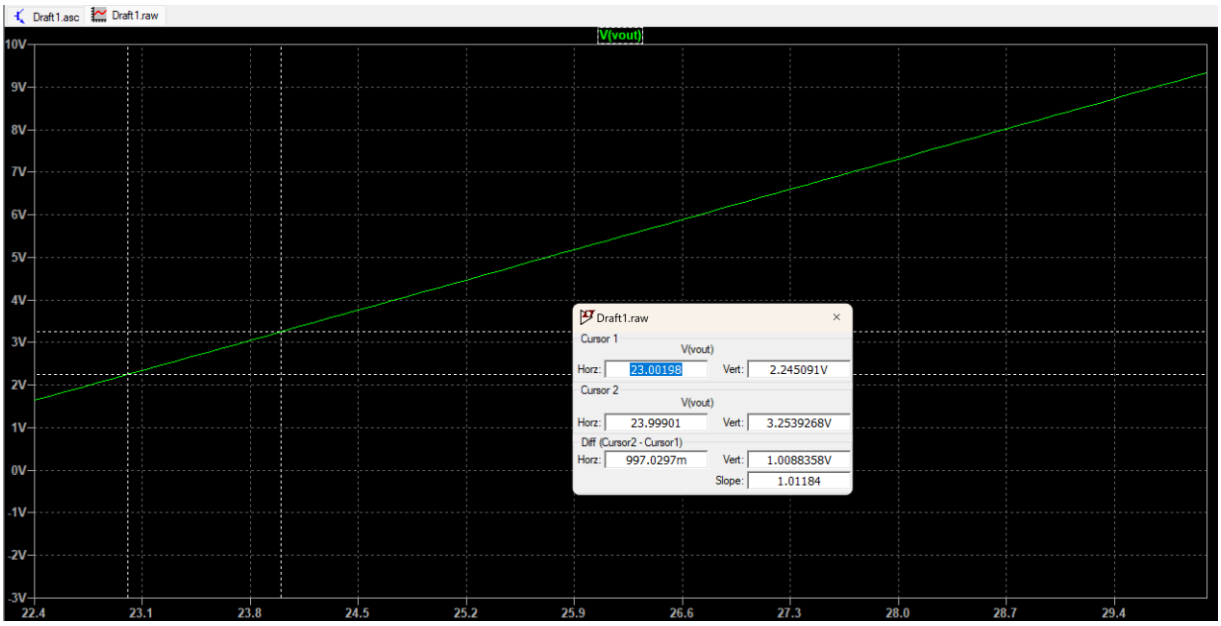


Figure 2.7: Vout change with 1 degree celcius difference between 23 and 24 degrees.

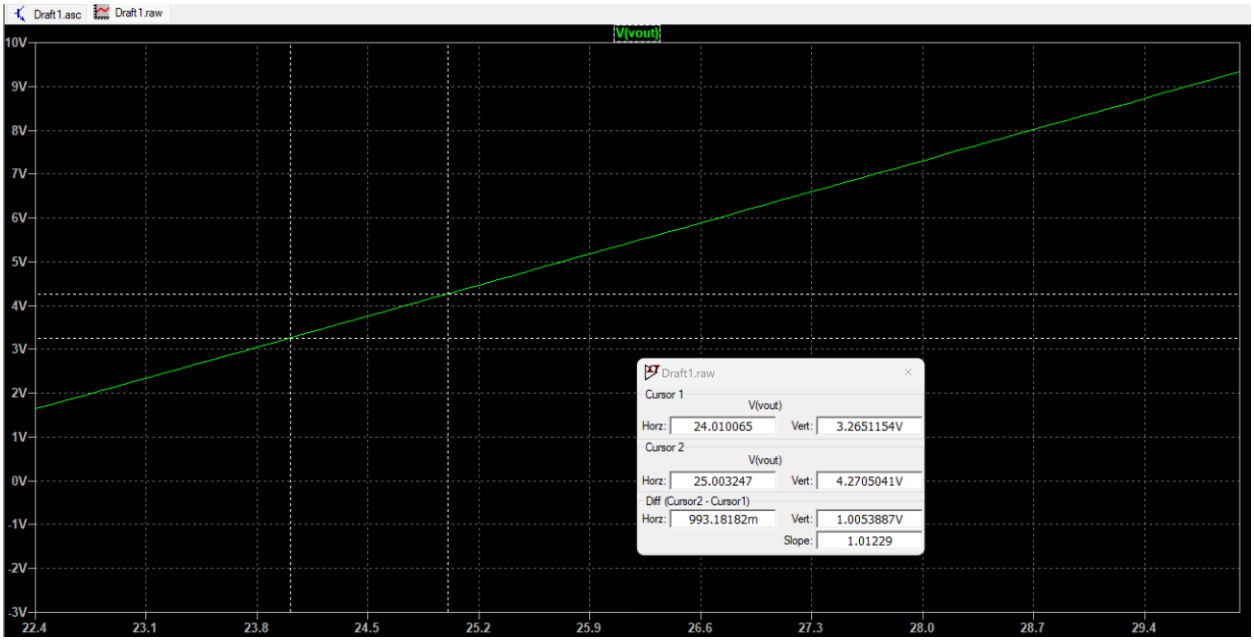


Figure 2.8: Vout change with 1 degree celcuis difference between 24 and 25 degrees.

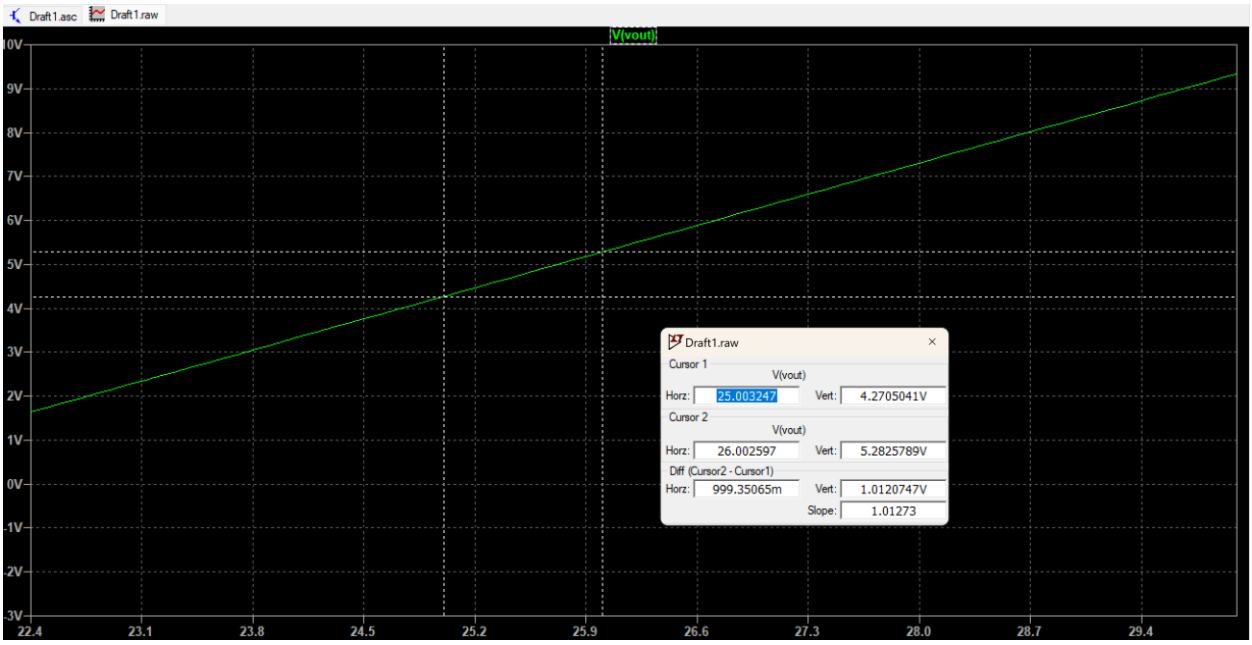


Figure 2.9: Vout change with 1 degree celcius difference between 25 and 26 degrees.

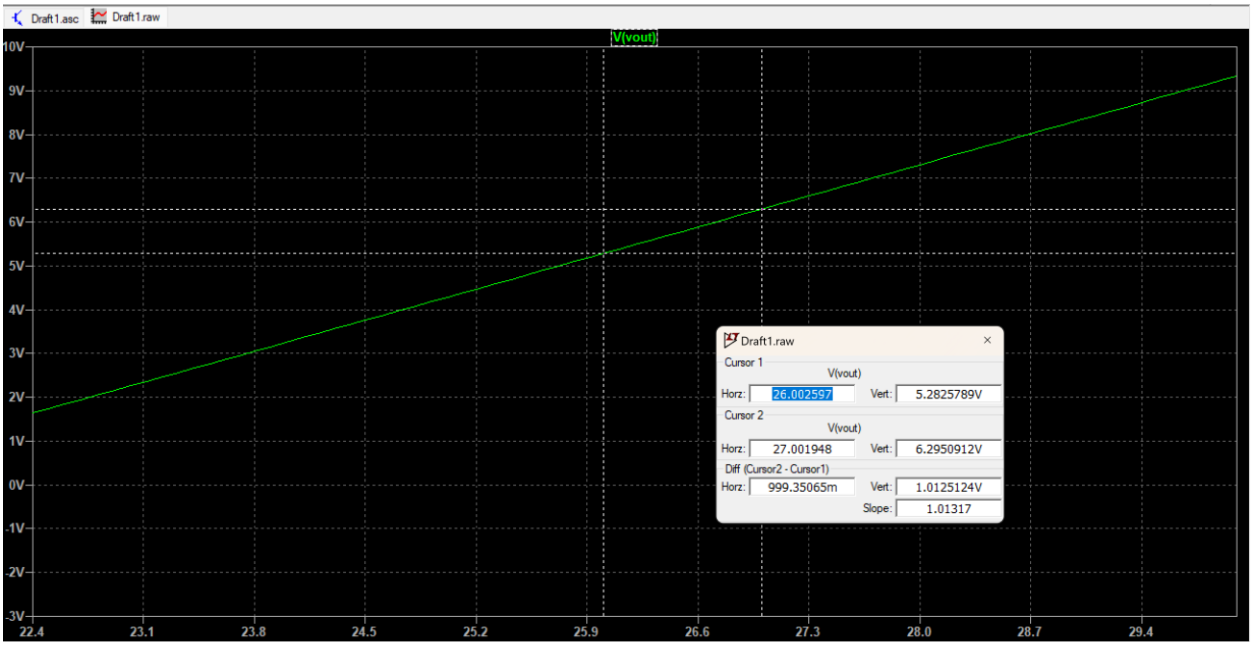


Figure 2.10: Vout change with 1 degree celcius difference between 26 and 27 degrees.

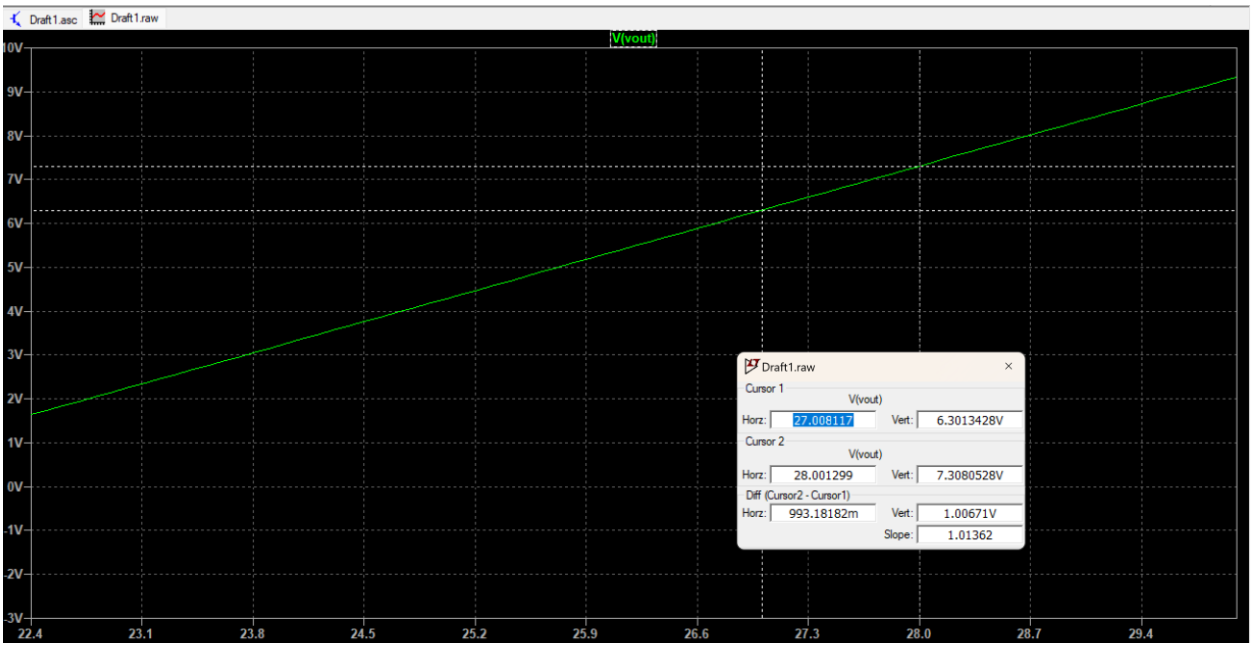


Figure 2.11: Vout change with 1 degree celcius difference between 27 and 28 degrees.

As it can be seen from all of the figures, the slopes of these plots are so close to 1, which indicates that the output voltage increases by 1V with each 1 degree celcius increase which satisfies our requirement for each interval.

3. A red LED should turn on when the sensor's temperature exceeds +5±0.5° C the room temperature.

When the current on the LED is measured, we expect to see that there will be a current flow then it exceeds the required temperature. Here on this plot, this can be seen easily:

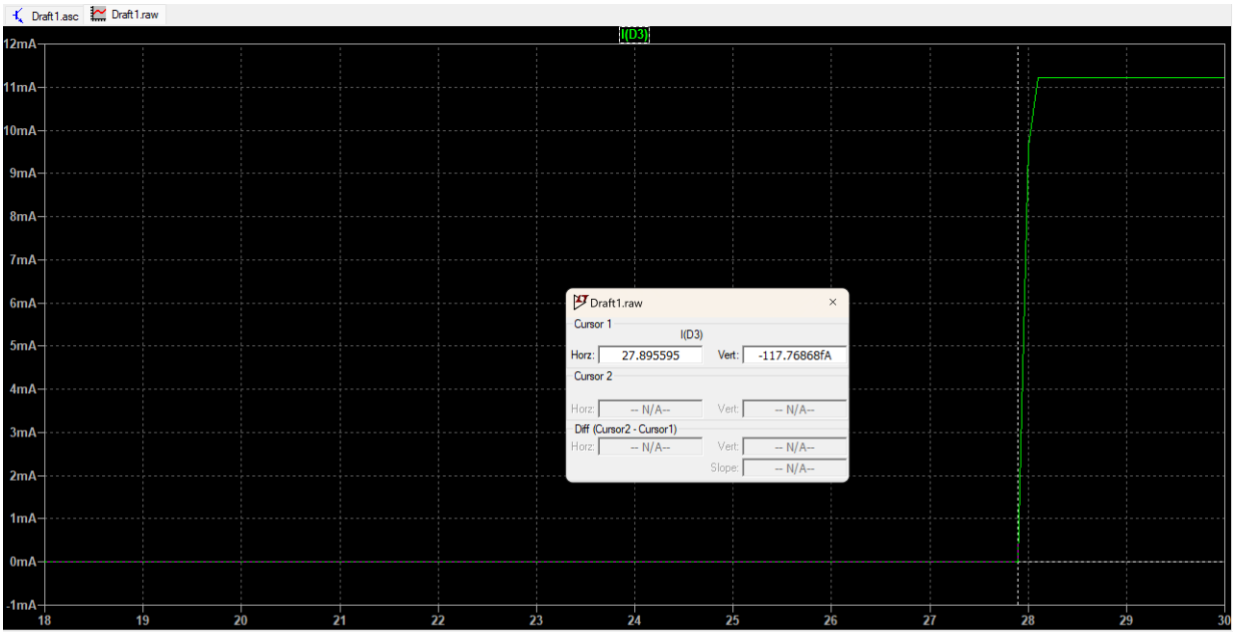


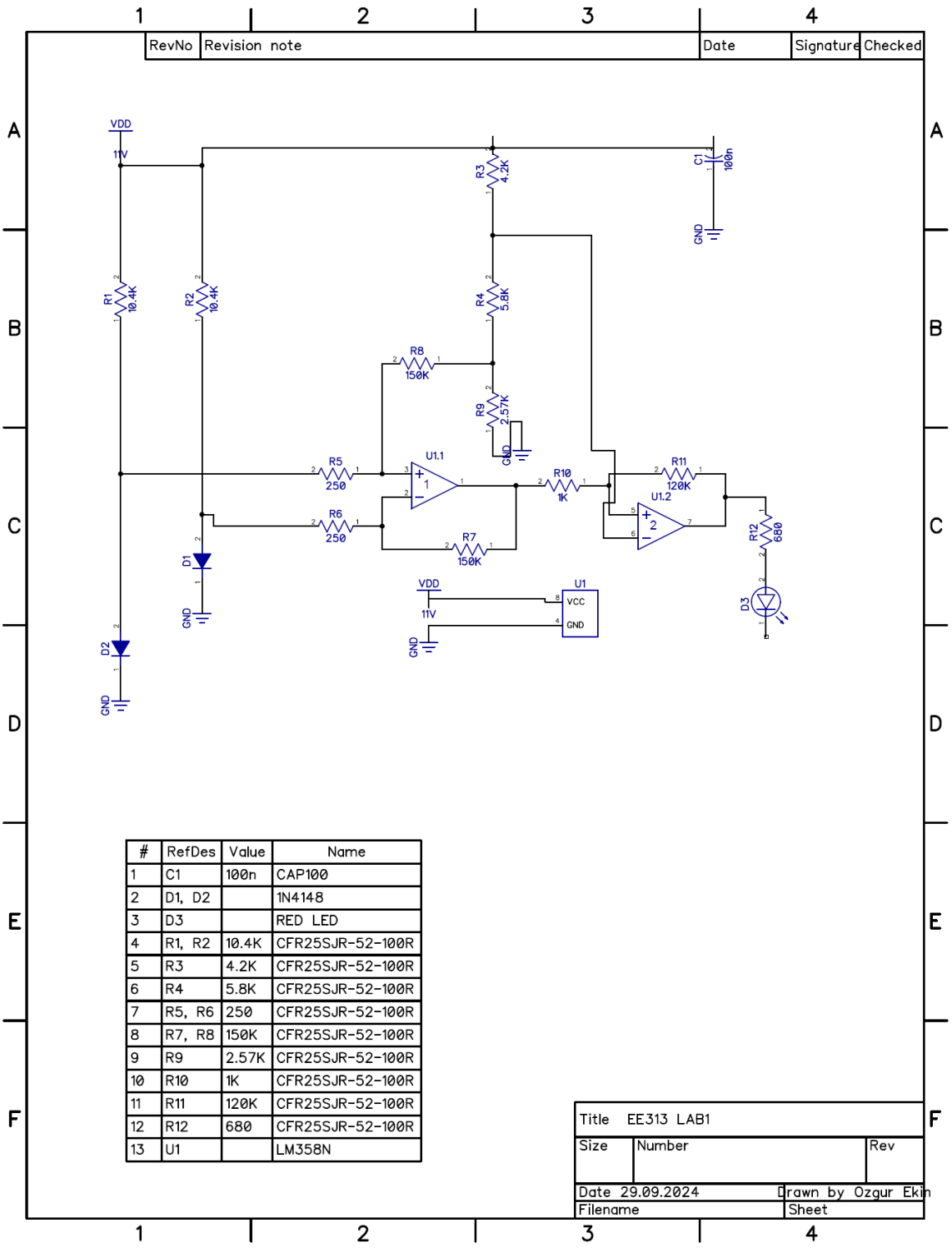
Figure 2.12: The current on the LED

As it can be seen on the figure, the current started to flow around 27.89 degrees celcius which satisfies our requirement of **+5±0.5°C the room temperature.**

4. **The LED should never flicker around the thresholds: It should have a 0.1° C (0.1V) hysteresis.**

By also looking at Figure 2.12, we can easily say that with 0.1 degrees below the turn-on point, the current of the LED goes to 0, which indicates that it is completely OFF.

Schematic:



#	RefDes	Value	Name
1	C1	100n	CAP100
2	D1, D2		1N4148
3	D3		RED LED
4	R1, R2	10.4K	CFR25SJR-52-100R
5	R3	4.2K	CFR25SJR-52-100R
6	R4	5.8K	CFR25SJR-52-100R
7	R5, R6	250	CFR25SJR-52-100R
8	R7, R8	150K	CFR25SJR-52-100R
9	R9	2.57K	CFR25SJR-52-100R
10	R10	1K	CFR25SJR-52-100R
11	R11	120K	CFR25SJR-52-100R
12	R12	680	CFR25SJR-52-100R
13	U1		LM358N

Figure 3: Schematic of the design.

HARDWARE PART:

Part A:

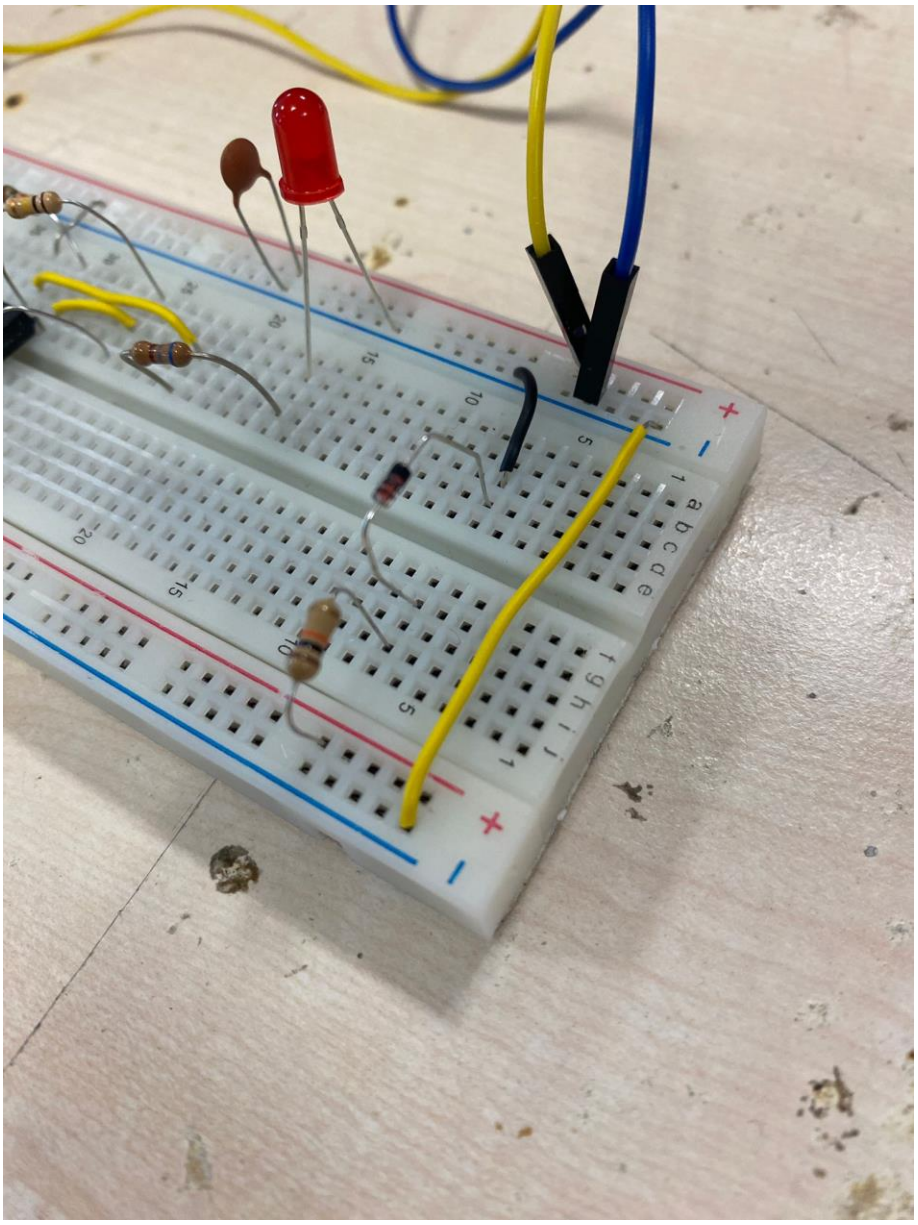


Figure 4: Part A, hardware design.

In this part, I built the required circuit to get the measurements and compare them with the calculations on the software part. Here a resistance of 10kΩ is used with a 10V voltage source. Then, a 1N4148 diode is added to the circuit to measure V_d and I_d values. Here are the measurements on the multimeter:

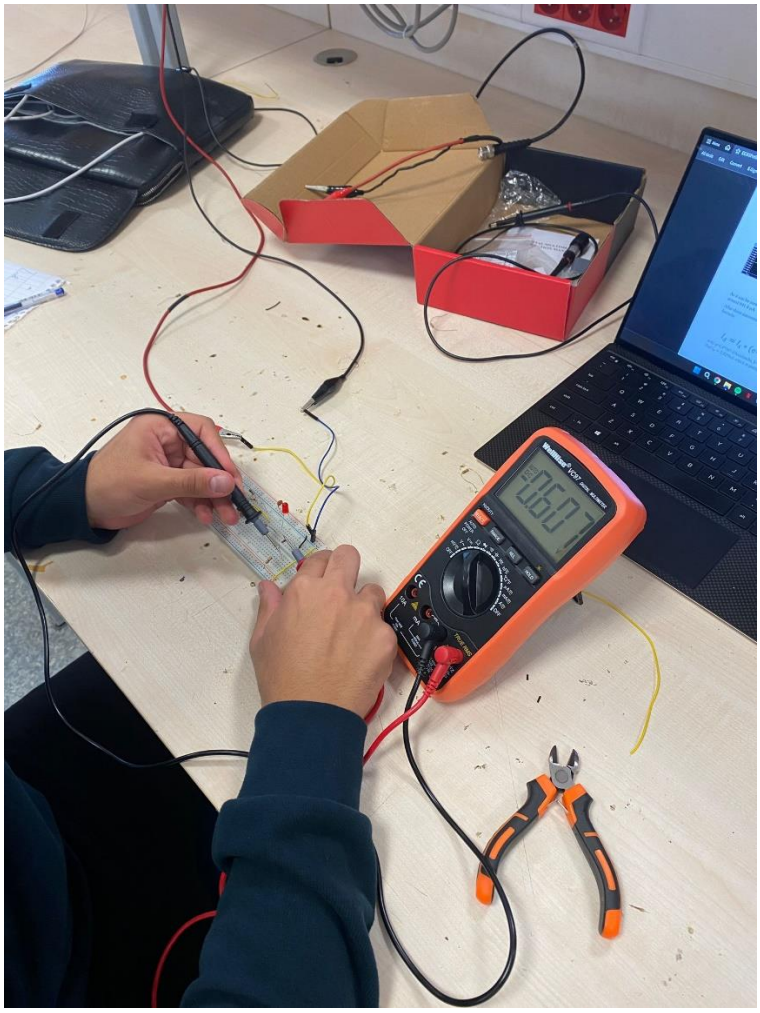


Figure 4.1: Measurement of V_d

On software we had the value of $V_d = 581.88mV$ and here, I measured $V_d = 607mV$. This corresponds to an error of **4.317%** which is a nice result.

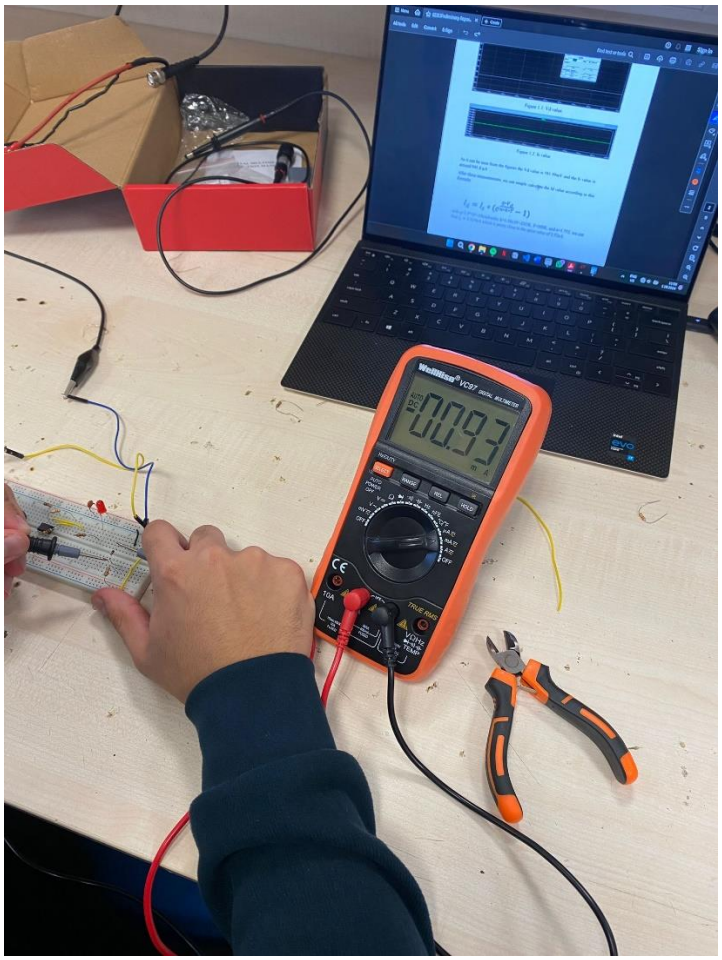


Figure 4.2: Measurement of I_d.

On software, the calculated value of the current was I_d=941.8µA and here, in the hardware measurement, this value is I_d=930µA which is again, a nice result with an error of **-1.2529%.**

Now to calculate the measured I_s value, we can use the same formula just like we used it to find the software result. The software result was $I_s \approx 2.502$nA and now, on the hardware, the result is $I_s \approx 1.903$nA. The error on these results is **-23.9408%** which is quite a large error. The main reason behind this error is the multiplication coefficients of the measurements we got. Even if we get good results on I_d *and* V_d , they are still not enough to get perfect results. The reason might be human error, or the varience of real components values due to our conditions on the lab. Overall, we still can measure and calculate the value of I_s.

Part B:

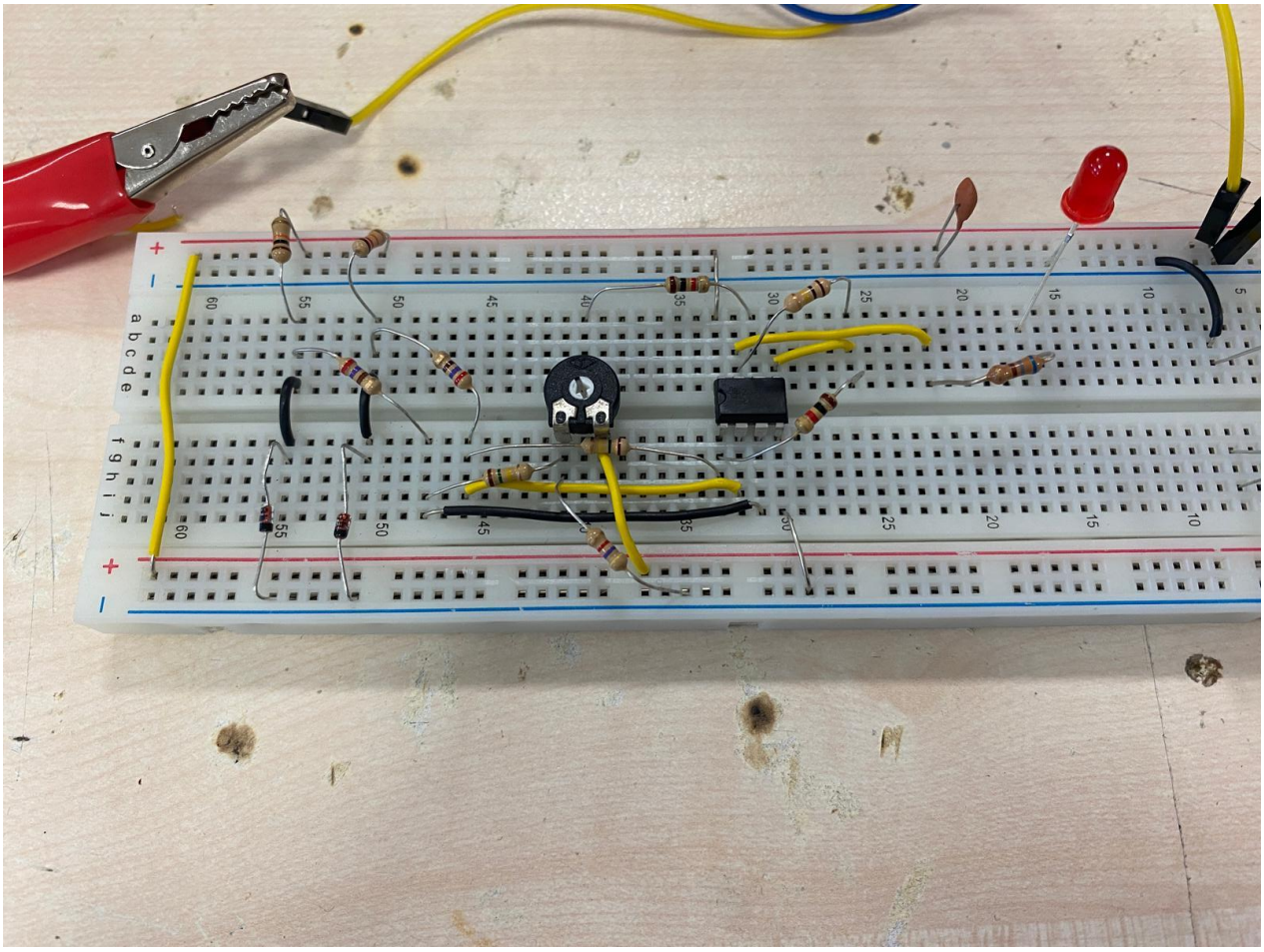


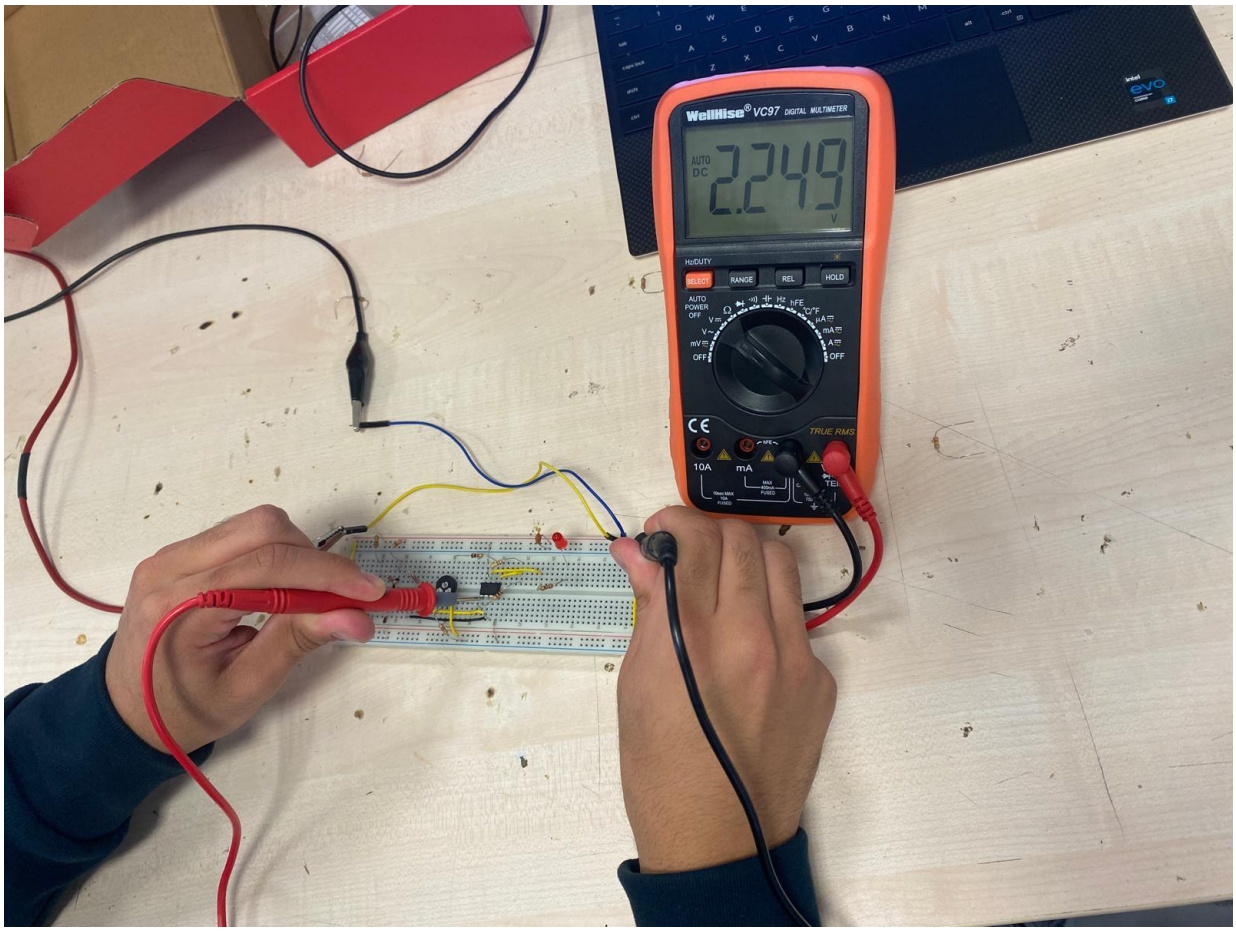
Figure 5: Part B, Differential Temperature Sensor Hardware Circuit

Here, in this part, the required circuit is built on hardware. Using Vcc=11V and all of the resistor values as closest to their standard values like,
1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2, the circuit is created and it is ready to be used. At first, the desired value of Vout, which was 2.25V in this case was lower in the measurements and therefore, the circuit does not respond the increase of the temperature as we want. So the value of $R_3 = 2.7K\Omega$ in the first case was increased to $R_3 = 3.9K\Omega$. This makes the value of V_R increase and thus, V_{out} according to the calculations on the formula below:

$$V_R = V_{CC} \frac{R_3}{R_3 + R_4 + R_5}$$

The values of R_4 and R_5 are adjusted by the potentiometer on the circuit according to the values calculated. To make the measurements right, the temperature is also measured with the multimeter on both cases. Also, the moment where the led just turns on is recorded. This is because the voltage increases really fast during the measurements and getting the right moment is important. Also, the value where the increase almost stops is measured too.

Here are the results and their comparisons with the software results:



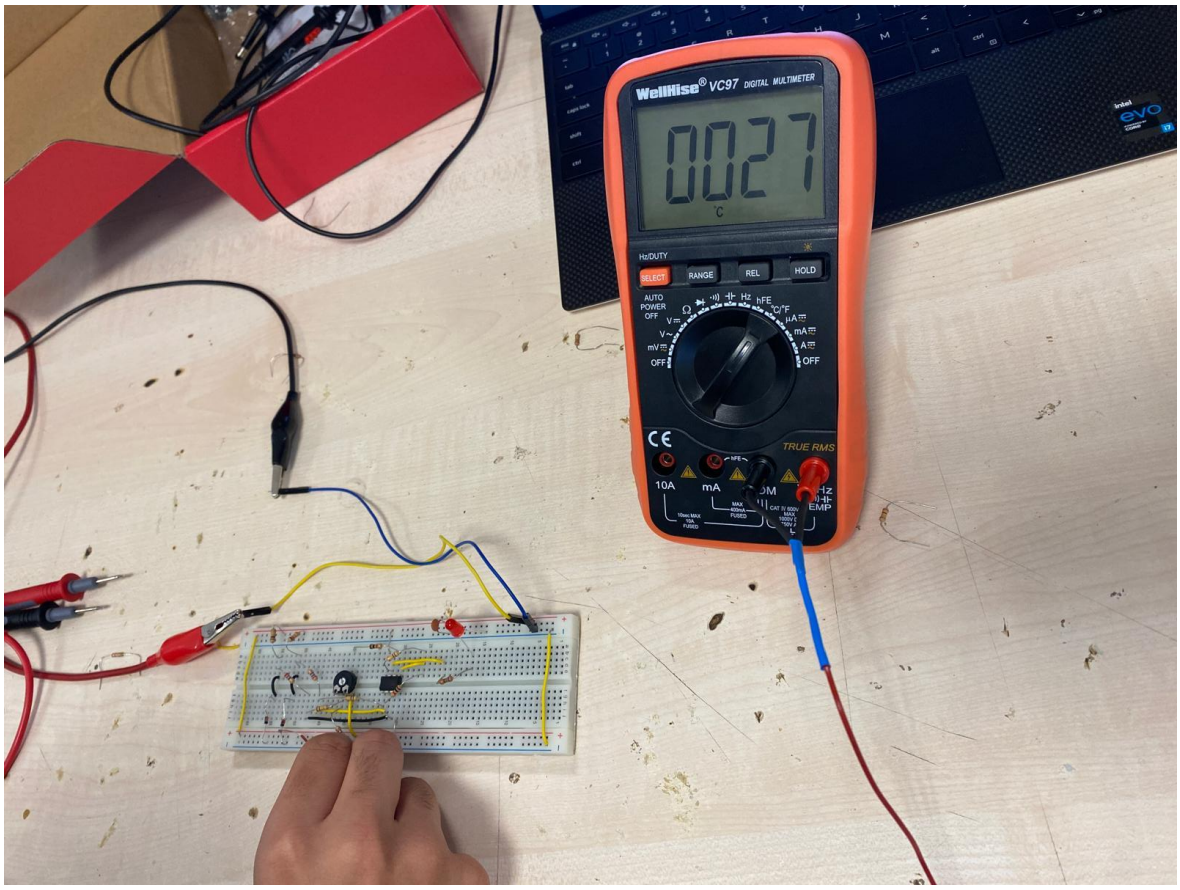


Figure 5.1: Measurement of V_{out} when both diodes are on room temperature.

As it can be seen on the images, when both diodes are at the room temperature, the LED is OFF. The value of V_{out}=2.249V which is almost perfect with an error of **-0.0444%**. The expected value is 2.25V according to the formula below when $V_{cc} = 11V$:

$$V_{out} = (V_{cc} - 2)/4$$

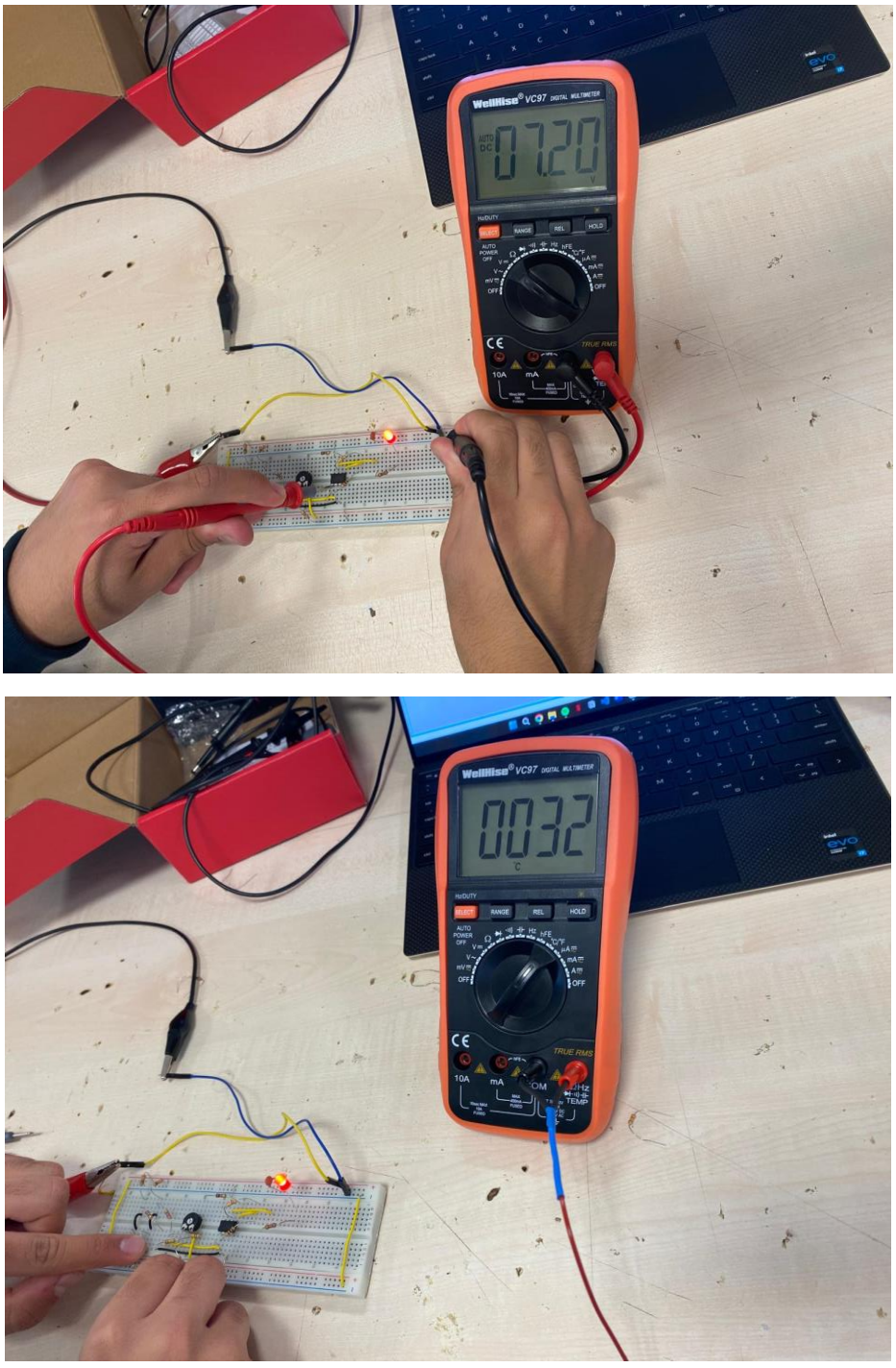


Figure 5.2: Measurement of V_{out} when temperature is increased to 32 degree Celcius

As it can be seen on the figures the LED turns ON just the moment when V_{out}=7.2V which is in range of the given requirements as follows: +5±0.5°C

When the increase is calculated, when it is 1 degree Celcius warmer, the voltage increases around 0.99V which is almost 1V and it is in the 10% tolerance range. Also, here is an image where the increase of temperature is almost done:

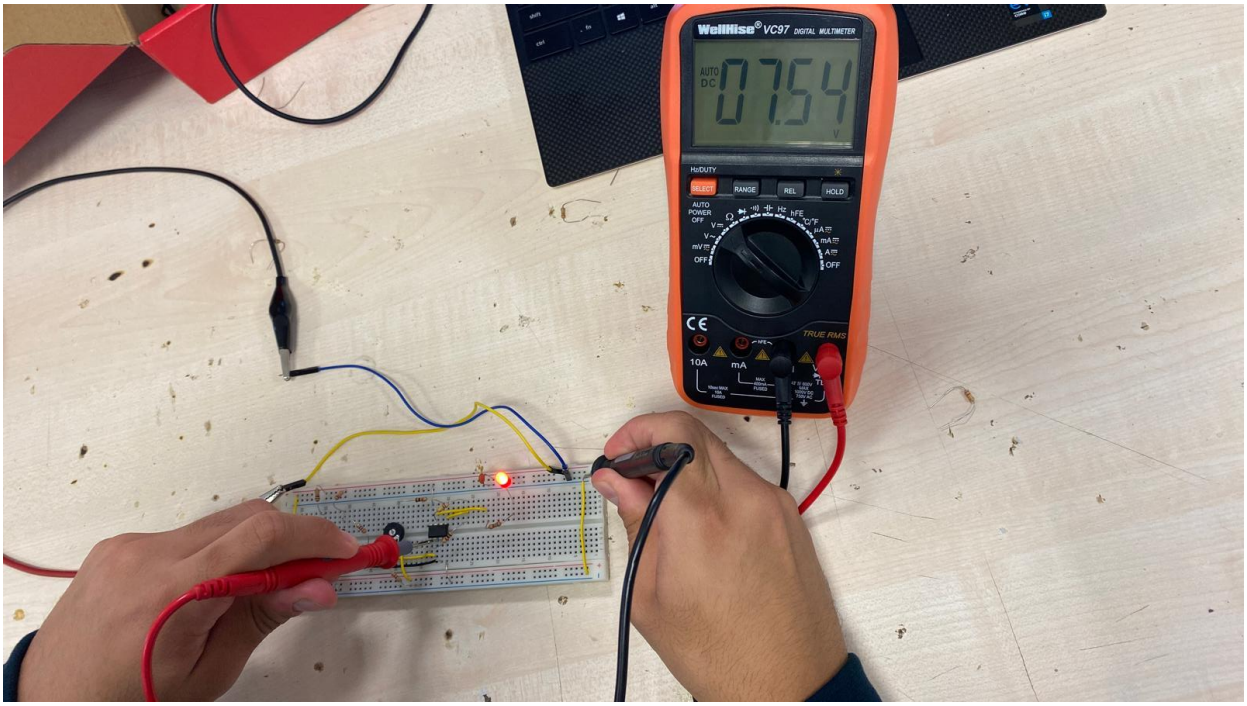


Figure 5.3: Extra measurement of V_{out}

This is the increase limit since I used my finger to warm up the diode. Higher values can be achieved using a lighter etc. to heat up the diode more.

Also, there is no flickering around the threshold values which satisfies the 4th requirement on out lab assignment.

Comments and Conclusion:

Overall, in this lab experiment, calculation and measurement of I_s is learned, and implemented on a hardware circuit. A differential temperature sensor is built on hardware and the results have been compared with the software results. To talk specifically on my experiment, I actually got really bad results at first place. The voltage gap on both diodes were really high at first place which resulted on a high voltage difference due to amplification. I switched to a different pair of diodes whose voltage values are really close which was important. Then, I got V_{out} pretty low. To debug this problem, I increased the value of R_3 which actually solved all

of my problems. This may be caused by the values of the components on the lab. Even the breadboard we use is changing the results we actually want. Human error is also a different but important factor on the errors. The adjustment of the potentiometer was also pretty hard. Out potentiometer given on this lab has a total resistance of 8kΩs but we were using 10kΩs of total resistance on the software. In the end, on the latest circuit, I did not even expected to get results this good on my hardware implementation. The results were almost the same as the results I got on my software calculations. Overall, this lab was pretty informative and also enjoyable when dealing with the diodes.